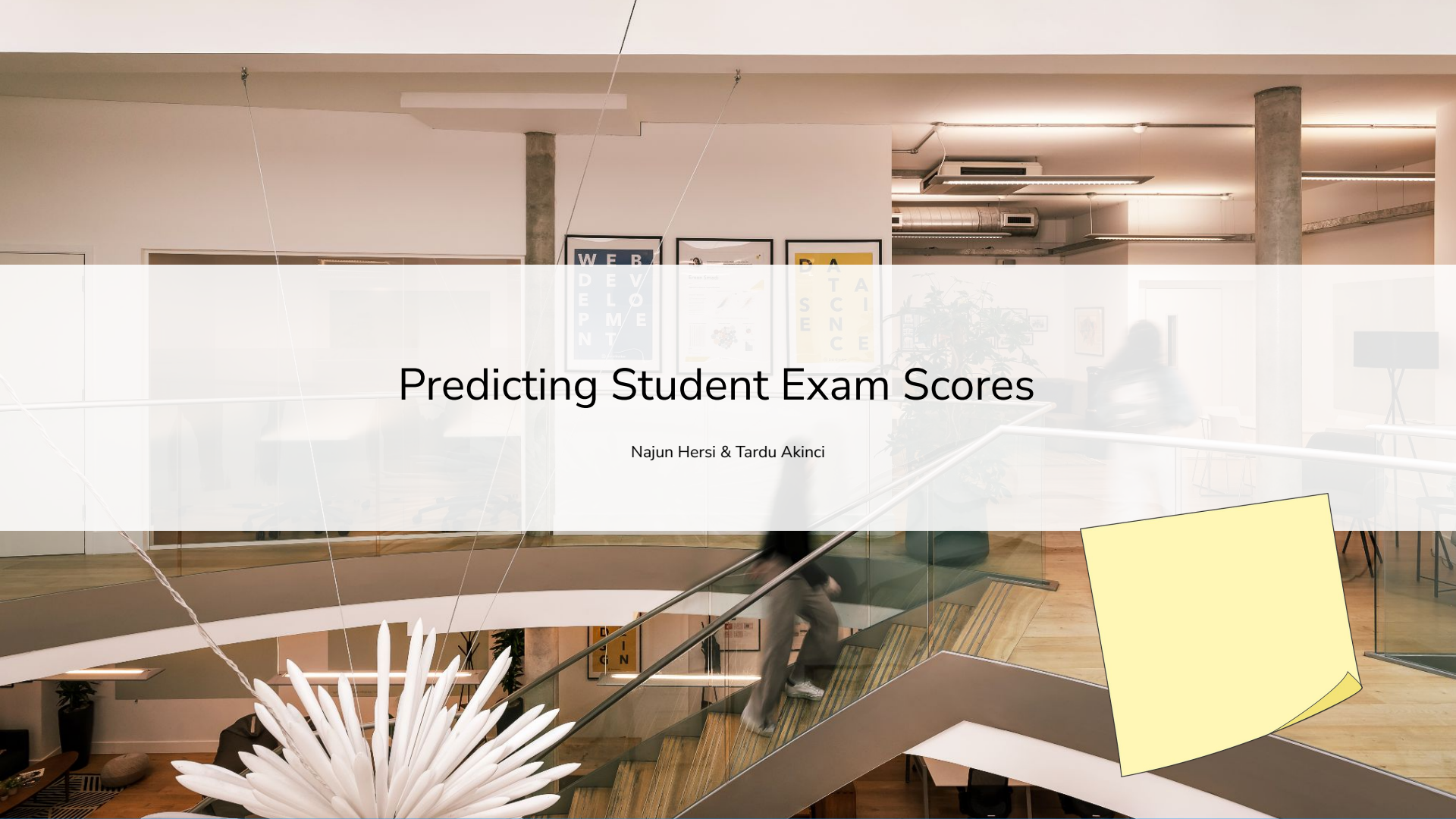


Predicting Student Exam Scores

Najun Hersi & Tardu Akinci

Agenda

- Problem Statement
- Hypothesis
- Dataset
- Data Cleaning
- Exploratory Data Analysis
- Model Building
- Model Validation
- Key Findings and Conclusions

The Problem Statement

Student academic performance depends on many different factors including study habits, attendance, and socio-economic conditions.

Understanding which variables have the strongest influence on exam performance can help educators improve student outcomes.

In this project, we analyze student data and build a machine learning model to predict final exam scores.

Hypothesis

We hypothesize that

- Hours studied
- Attendance
- Previous

will be the strongest predictors of a student's final exam score.

Data and Tools Used

Dataset: **Student Performance Factors**

- Source : Kaggle
- ~6600 student observations
- Each row represents 1 student

Dataset includes:

- Study habits
- Academic performance history
- Environmental and socio-economic variables

Goal: predict **Final Exam Score**

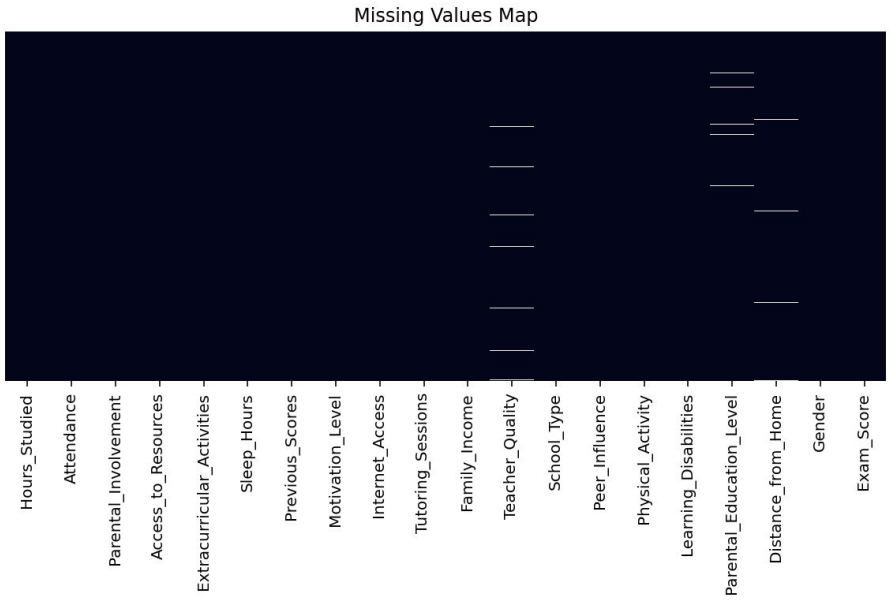
Tools used:

Python, Pandas, Numpy, Matplotlib/Seaborn, Scikit-learn

Data Cleaning

Steps performed:

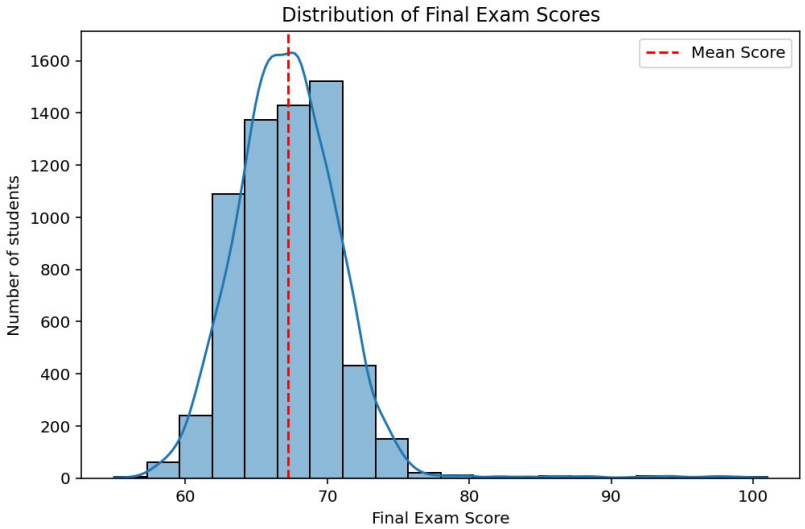
- Checked dataset structure and data types
- Identified missing values
- Visualized missing data using a heatmap
- Removed rows containing missing values
- Converted categorical variables using **one-hot encoding**



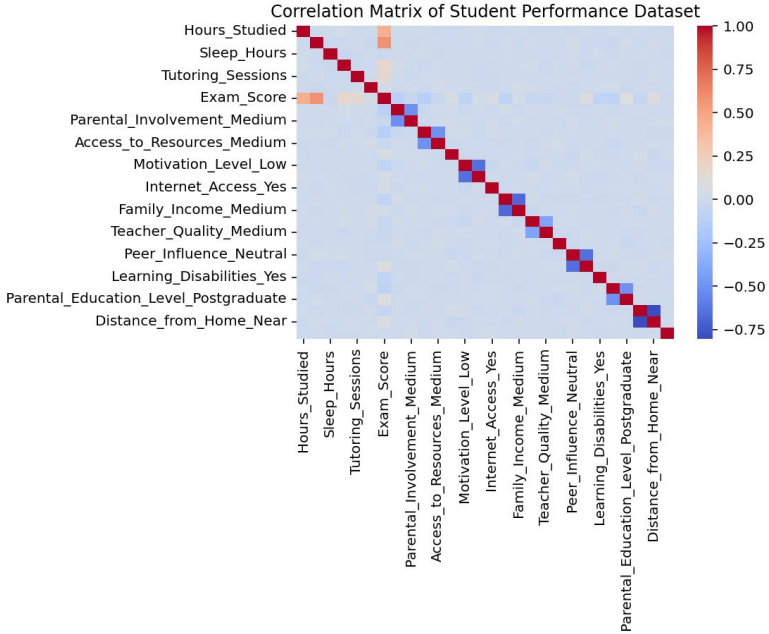
Result:

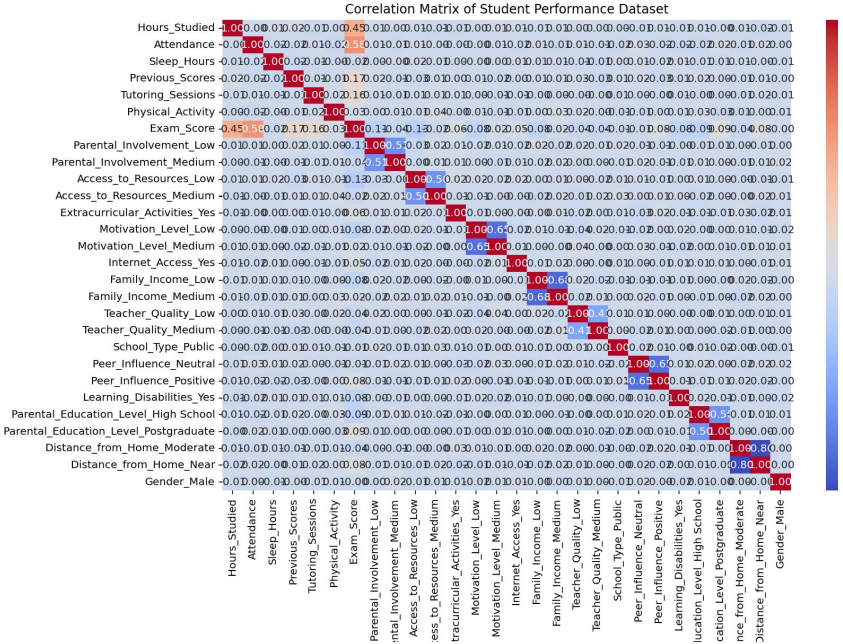
Clean dataset with **numerical variables ready for modeling**

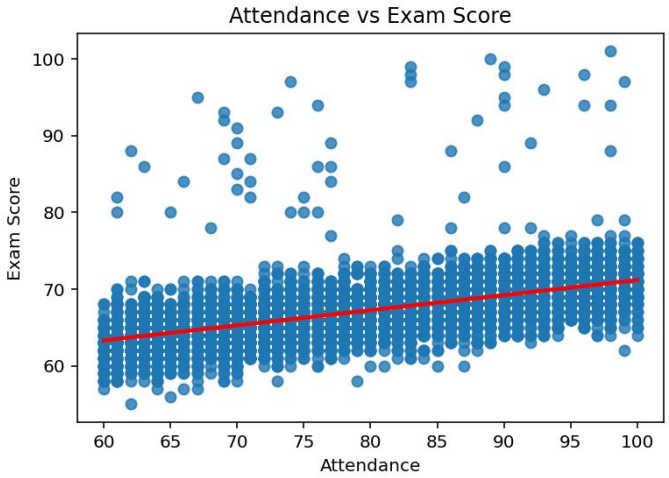
Exploratory Data Analysis

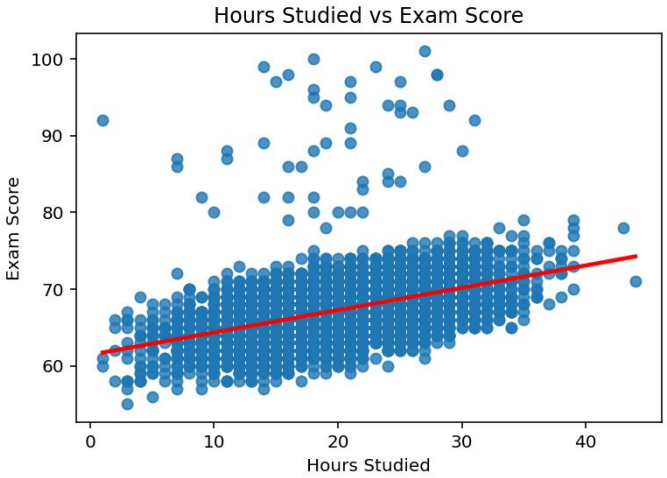


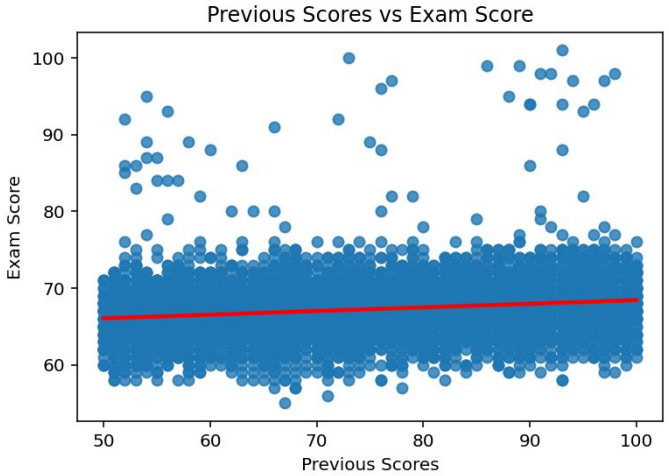
- Mean exam score: 67
- Distribution is approximately normal


Correlation Matrix of Student Performance Dataset
Hours_Studied
Sleep_Hours
Tutoring_Sessions
Exam_Score
Parental_Involvement_Medium
Access_to_Resources_Medium
Motivation_Level_Low
Internet_Access_Yes
Family_Income_Medium
Teacher_Quality_Medium
Peer_Influence_Neutral
Learning_Disabilities_Yes
Parental_Education_Level_Postgraduate
Distance_from_Home_Near
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75


Correlation Matrix of Student Performance Dataset
Hours_Studied
Attendance
Sleep_Hours
Previous_Scores
Tutoring_Sessions
Physical_Activity
Exam_Score
Parental_Involvement_Low
Parental_Involvement_Medium
Access_to_Resources_Low
Access_to_Resources_Medium
Extracurricular_Activities_Yes
Motivation_Level_Low
Motivation_Level_Medium
Internet_Access_Yes
Family_Income_Low
Family_Income_Medium
Teacher_Quality_Low
Teacher_Quality_Medium
School_Type_Public
Peer_Influence_Neutral
Peer_Influence_Positive
Learning_Disabilities_Yes
Parental_Education_Level_High School
Parental_Education_Level_Postgraduate
Distance_from_Home_Moderate
Distance_from_Home_Near
Gender_Male


Attendance vs Exam Score
Exam Score
Attendance
100
90
80
70
60
60
65
70
75
80
85
90
95
100


Hours Studied vs Exam Score
Exam Score
Hours Studied
100
90
80
70
60
0
10
20
30
40


Previous Scores vs Exam Score
Exam Score
Previous Scores
100
90
80
70
60
50
60
70
80
90
100

Statistical Analysis

- Descriptive statistics were used to summarize the dataset
- Correlation analysis was performed to examine relationships between variables
- Scatter plots were used to visualize trends in the data
- Linear regression was used to measure the impact of predictors on exam scores
- Formal hypothesis testing was not performed in this analysis

Descriptive statistics were used to summarize the dataset

Correlation analysis was performed to examine relationships between variables

Scatter plots were used to visualize trends in the data

Linear regression was used to measure the impact of predictors on exam scores

Formal hypothesis testing was not performed in this analysis

Modeling

Model used: Linear Regression

Why Linear Regression:

- Target variable (Exam_Score) is continuous
- Allows us to measure relationships between predictors and exam scores

Training Process:

- Predictor variables were selected from the cleaned dataset
- Data split into 80% training data and 20% testing data

Model training:

- Model trained using the training dataset
- Predictions generated for the test dataset

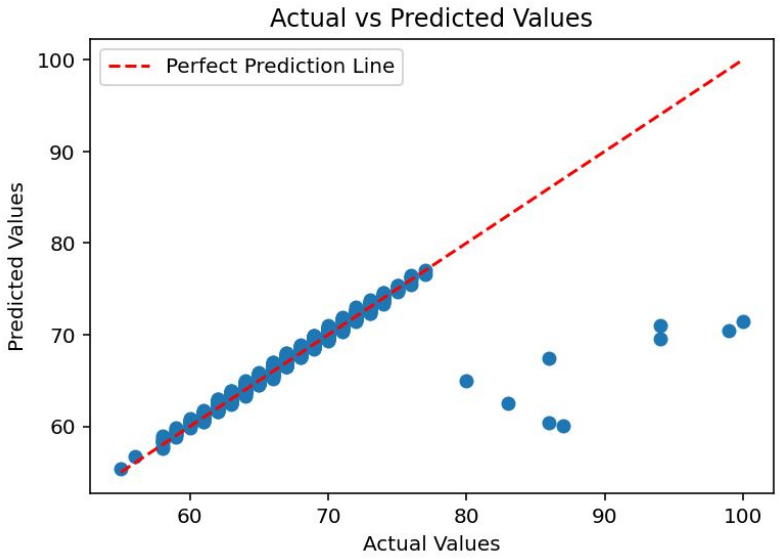
Model Validation

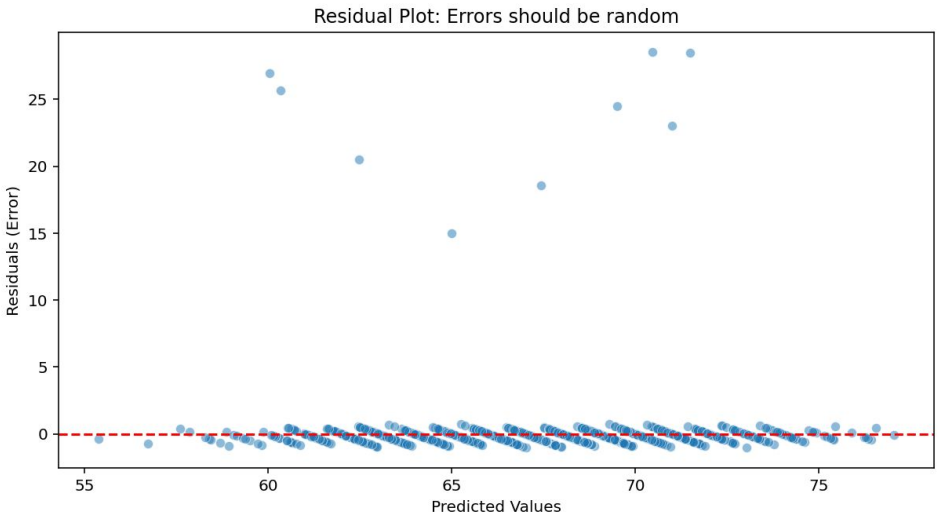
Metrics:

- R^2 Score ≈ 0.73
- Mean Absolute Error (MAE) ≈ 0.48
- Root Mean Squared Error (RMSE) ≈ 2.0

Cross validation:

- 5-fold cross validation used
- Mean R^2 ≈ **0.72**
- Confirms model performance is stable across different data splits


Actual vs Predicted Values
Perfect Prediction Line
Predicted Values
Actual Values
100
90
80
70
60
60
70
80
90
100


Residual Plot: Errors should be random
Residuals (Error)
Predicted Values
25
20
15
10
5
0
55
60
65
70
75

Summary Conclusions

- The linear regression model successfully predicted student exam scores.
- The model explained about **73% of the variation in exam scores ($R^2 \approx 0.73$)**.
- **Hours studied, attendance, and previous scores** were the strongest predictors of exam performance.
- Study effort and engagement play a major role in student success.
- Our hypothesis was **supported by the results of the analysis**.

Key Learnings

- Data cleaning is an essential step before building a machine learning model.
- Exploratory data analysis helps reveal patterns and relationships in the data.
- Linear regression can be used to quantify how different variables influence an outcome.
- Model validation techniques such as residual analysis and cross-validation help evaluate model reliability.
- Data science projects require both **technical analysis and interpretation of results**.

Thank You

References

- Kaggle – Student Performance Factors Dataset
- Python Scikit-learn Documentation
- Pandas Documentation
- Matplotlib / Seaborn Documentation